Condensate & Feedwater

GE BWR/4 Technology Course

R-304B – 2.6

Objectives

1. Identify the purposes of the Condensate and Feedwater system.
2. Recognize the purpose, function and operation of the following Condensate and Feedwater system major components:
   a. Condensate Storage Tank
   b. Main Condenser
   c. Main Condenser Hotwell
   d. Condensate Pumps
   e. Steam Jet Air Ejector Condenser
   f. Steam Packing Exhauster
   g. Condensate Demineralizers
   h. Condensate Booster Pumps
   i. Low Pressure Heaters
   j. Feedwater Pumps
   k. High Pressure Heaters

Objectives

3. Describe the following flowpaths of the Condensate and Feedwater system:
    a. Condensate & Demineralized Makup Water System
    b. Hotwell Level Control
    c. Condensate System
    d. Feedwater System
    e. Exhaust Hood Spray
    f. Short Cycle Cleanup
    g. Long Cycle Cleanup

Objectives

4. Identify the purpose or function of the following Condensate and Feedwater System setpoints:
    a. Condensate Demineralizer High Temperature
    b. Condensate Demineralizer Resin Depletion
    c. Condensate Booster Pump Low Suction Pressure
    d. Feedwater Pump Low Suction Pressure

Objectives

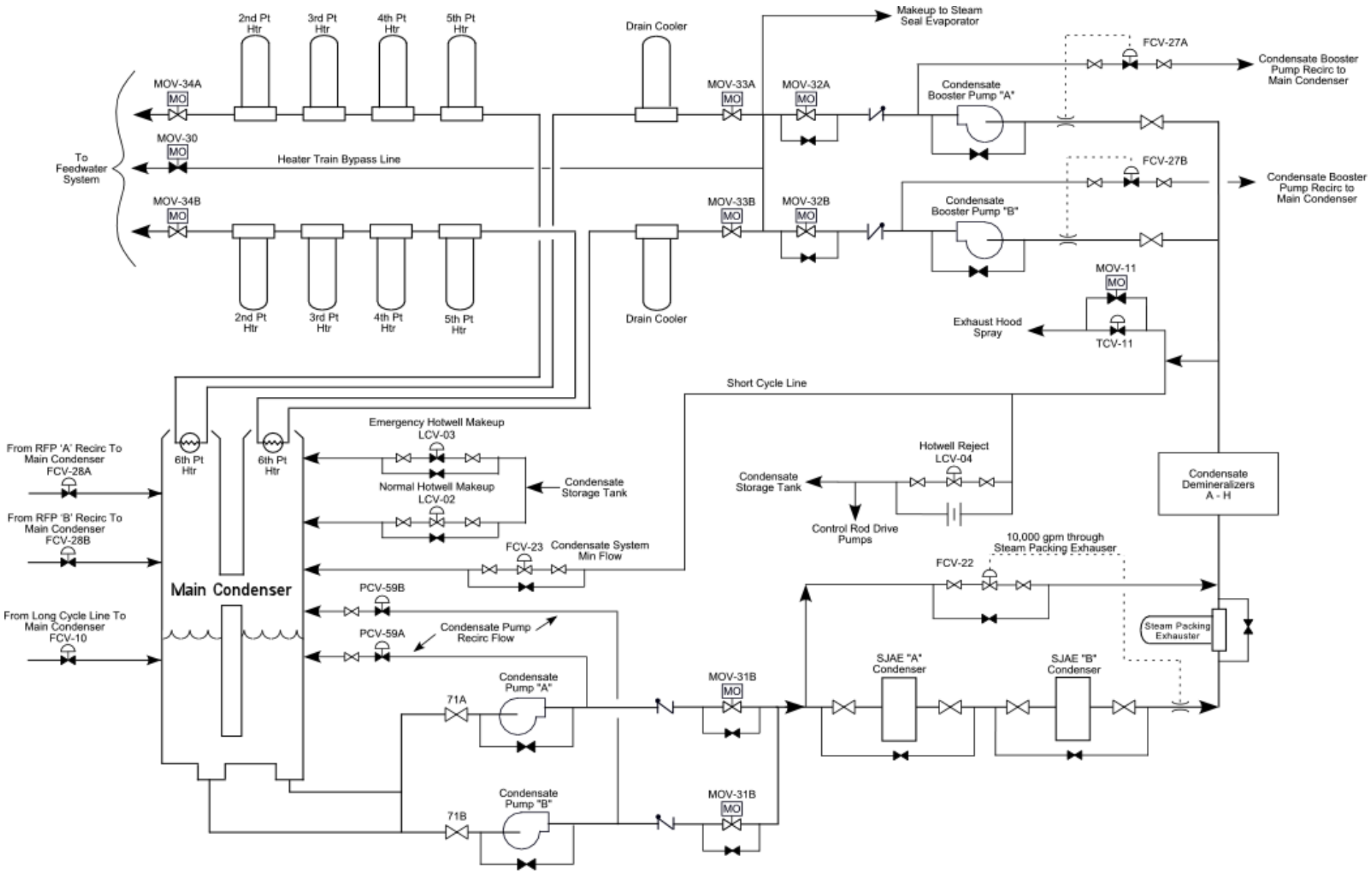
5. Describe how the Condensate and Feedwater system interrelates with the following systems/components:
    a. Circulating Water System
    b. Control Rod Drive System
    c. Feedwater Control System
    d. Reactor Water Cleanup System
    e. High Pressure Coolant Injection System
    f. Reactor Core Isolation Cooling System
    g. Main Steam System
    h. Offgas System

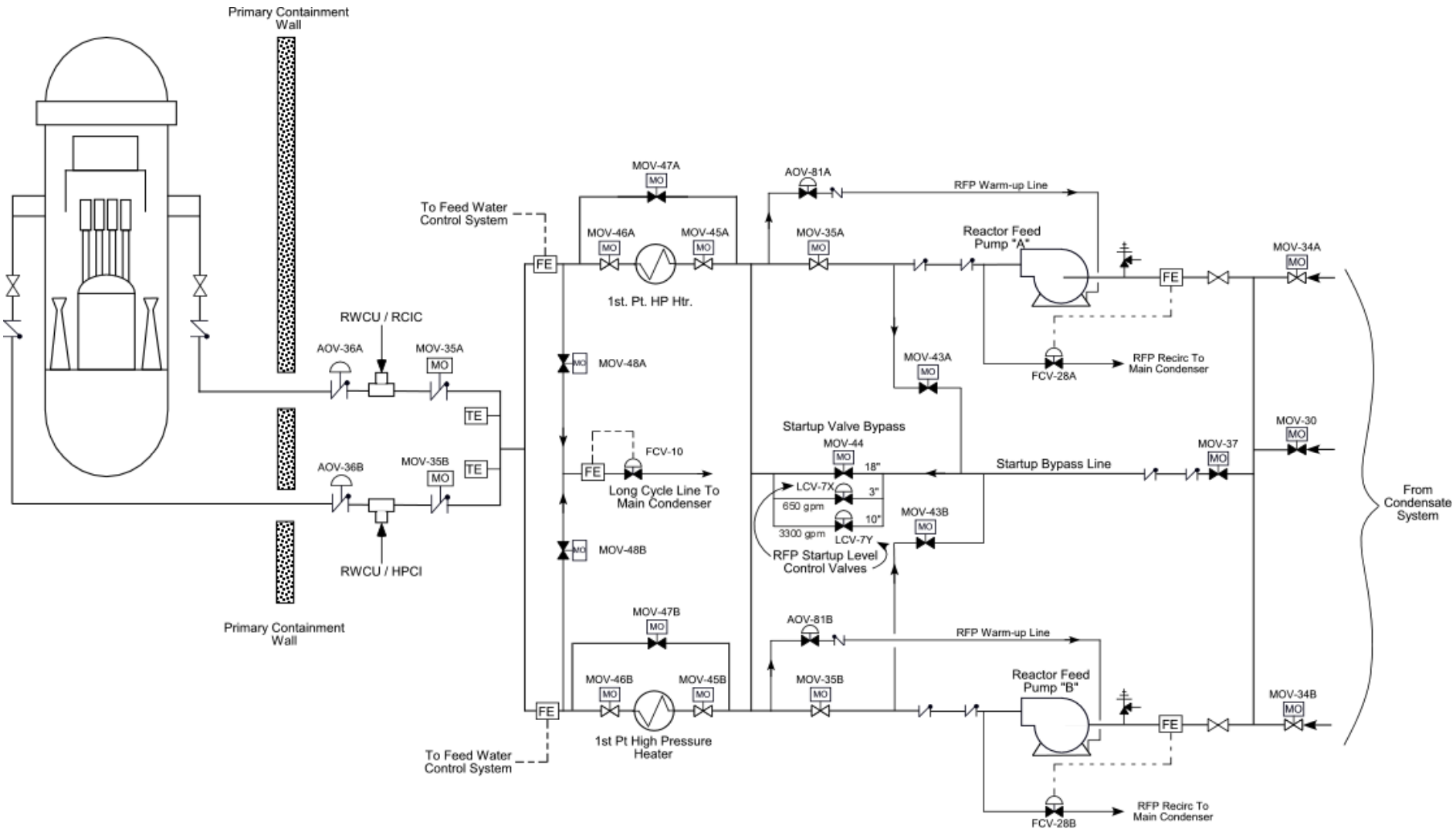
Condensate & Feedwater System Purposes

- Condense Steam
- Collect Steam Drains
- Remove Noncondensible Gases
- Purify
- Preheat
- Pump Water to Reactor Vessel
- Provide path to Reactor Vessel for :
  - High Pressure Coolant Injection
  - Rector Core Isolation Cooling
  - Reactor Water Cleanup

Condensate & Feedwater System Major Components

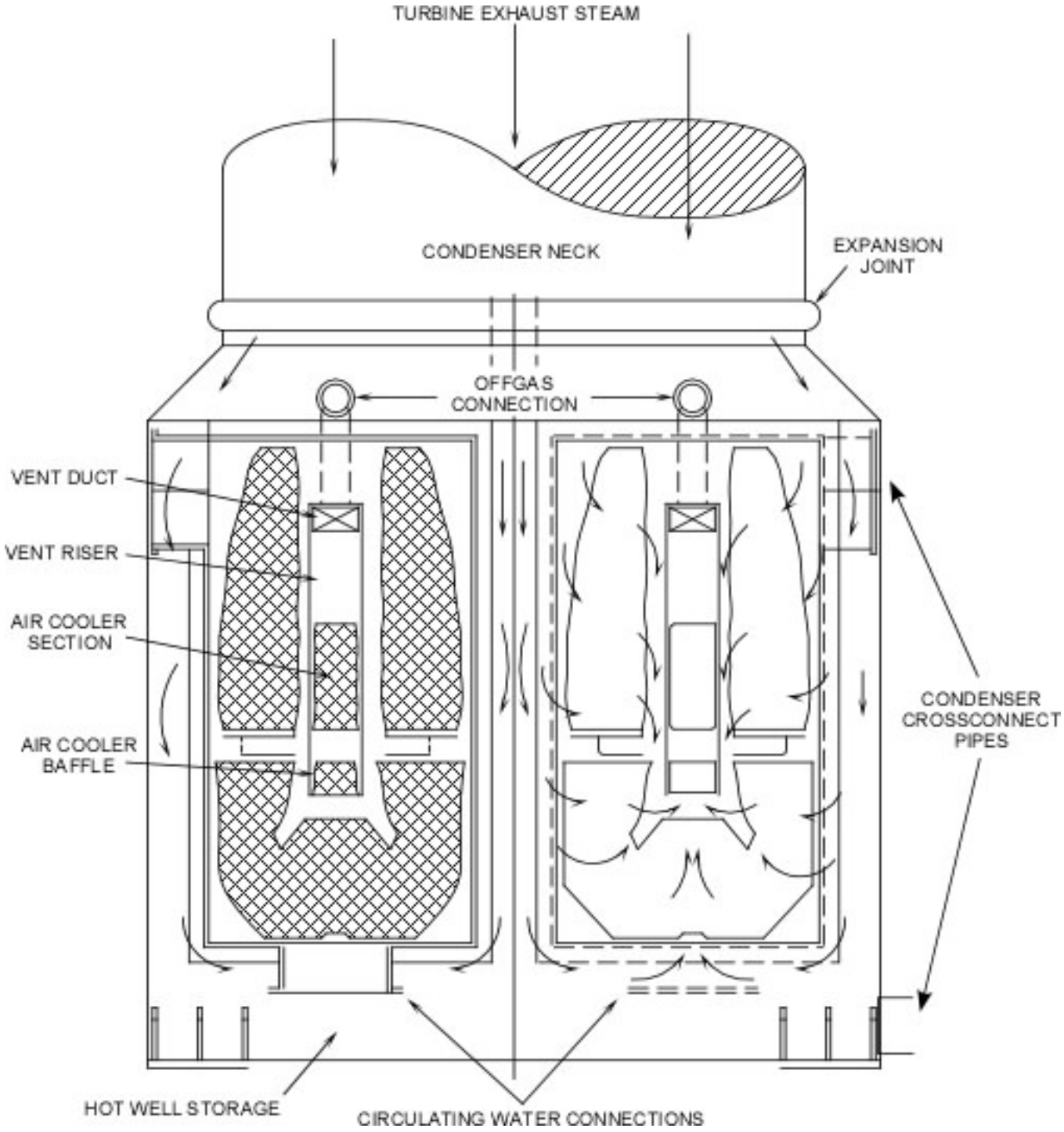
- Condensate Storage Tank
- Main Condenser
- Main Condenser Hotwell
- Condensate Pumps
- Steam Jet Air Ejector Condenser
- Steam Packing Exhauster
- Condensate Demineralizers
- Condensate Booster Pumps
- Low Pressure Heaters
- Feedwater Pumps
- High Pressure Heaters
- Extraction Steam


2nd Pt Htr
3rd Pt Htr
4th Pt Htr
5th Pt Htr
Drain Cooler
Makeup to Steam Seal Evaporator
FCV-27A
Condensate Booster Pump Recirc to Main Condenser
MOV-34A
MO
MOV-33A
MOV-32A
Condensate Booster Pump "A"
MOV-30
To Feedwater System
Heater Train Bypass Line
FCV-27B
Condensate Booster Pump Recirc to Main Condenser
MOV-34B
MOV-33B
MOV-32B
Condensate Booster Pump "B"
MOV-11
Exhaust Hood Spray
TCV-11
Short Cycle Line
Emergency Hotwell Makeup LCV-03
Normal Hotwell Makeup LCV-02
Condensate Storage Tank
Hotwell Reject LCV-04
Condensate Demineralizers A - H
From RFP 'A' Recirc To Main Condenser FCV-28A
From RFP 'B' Recirc To Main Condenser FCV-28B
From Long Cycle Line To Main Condenser FCV-10
6th Pt Htr
Main Condenser
FCV-23
Condensate System Min Flow
Control Rod Drive Pumps
10,000 gpm through Steam Packing Exhauser
FCV-22
PCV-59B
PCV-59A
Condensate Pump Recirc Flow
Steam Packing Exhauster
SJAE "A" Condenser
SJAE "B" Condenser
Condensate Pump "A"
71A
MOV-31B
Condensate Pump "B"
71B


Primary Containment Wall
RWCU / RCIC
AOV-36A
MOV-35A
MO
TE
AOV-36B
MOV-35B
RWCU / HPCI
Primary Containment Wall
To Feed Water Control System
FE
MOV-47A
MOV-46A
MOV-45A
1st. Pt. HP Htr.
MOV-48A
FCV-10
Long Cycle Line To Main Condenser
MOV-48B
MOV-47B
MOV-46B
MOV-45B
1st Pt High Pressure Heater
To Feed Water Control System
AOV-81A
RFP Warm-up Line
MOV-35A
Reactor Feed Pump "A"
MOV-43A
FCV-28A
RFP Recirc To Main Condenser
Startup Valve Bypass
MOV-44
18"
LCV-7X
3"
650 gpm
10"
3300 gpm
LCV-7Y
RFP Startup Level Control Valves
MOV-43B
Startup Bypass Line
MOV-37
MOV-30
MOV-34A
From Condensate System
AOV-81B
RFP Warm-up Line
MOV-35B
Reactor Feed Pump "B"
FCV-28B
RFP Recirc To Main Condenser
MOV-34B

Main Condenser

- Two Condensers
- Deaerate
- Radial Flow Surface Condensers
- Divided Water Boxes
- Collects LP Turbine Exhaust
- Cooled by Circulating Water System
- Heat Sink for:
  - RFP Exhaust
  - Turbine Bypass
  - Low Pressure Htr Drains
  - Air Ejector Condenser Drains
  - Steam Packing Exhauster
  - Feedwater Htr shell vents
  - Pump flows
  - Minimum Recirculation Flow


TURBINE EXHAUST STEAM
CONDENSER NECK
EXPANSION JOINT
OFFGAS CONNECTION
VENT DUCT
VENT RISER
AIR COOLER SECTION
AIR COOLER BAFFLE
CONDENSER CROSSCONNECT PIPES
HOT WELL STORAGE
CIRCULATING WATER CONNECTIONS

Main Condenser Hotwell

- Integral part of the Main Condenser
- Collects condensate
- Connected by Equalizing lines
- Designed to retain water for minimum of two minutes
  - Nitrogen – 16 decay half-life - 7.11 seconds
  - At minimum operating hotwell levels

Condensate Pumps

- Two pumps
- Provide adequate net positive suction head for Condensate Booster Pumps
- Capacity 12,153 gpm at ~ 200 psig

Steam Jet Air Ejector Condenser

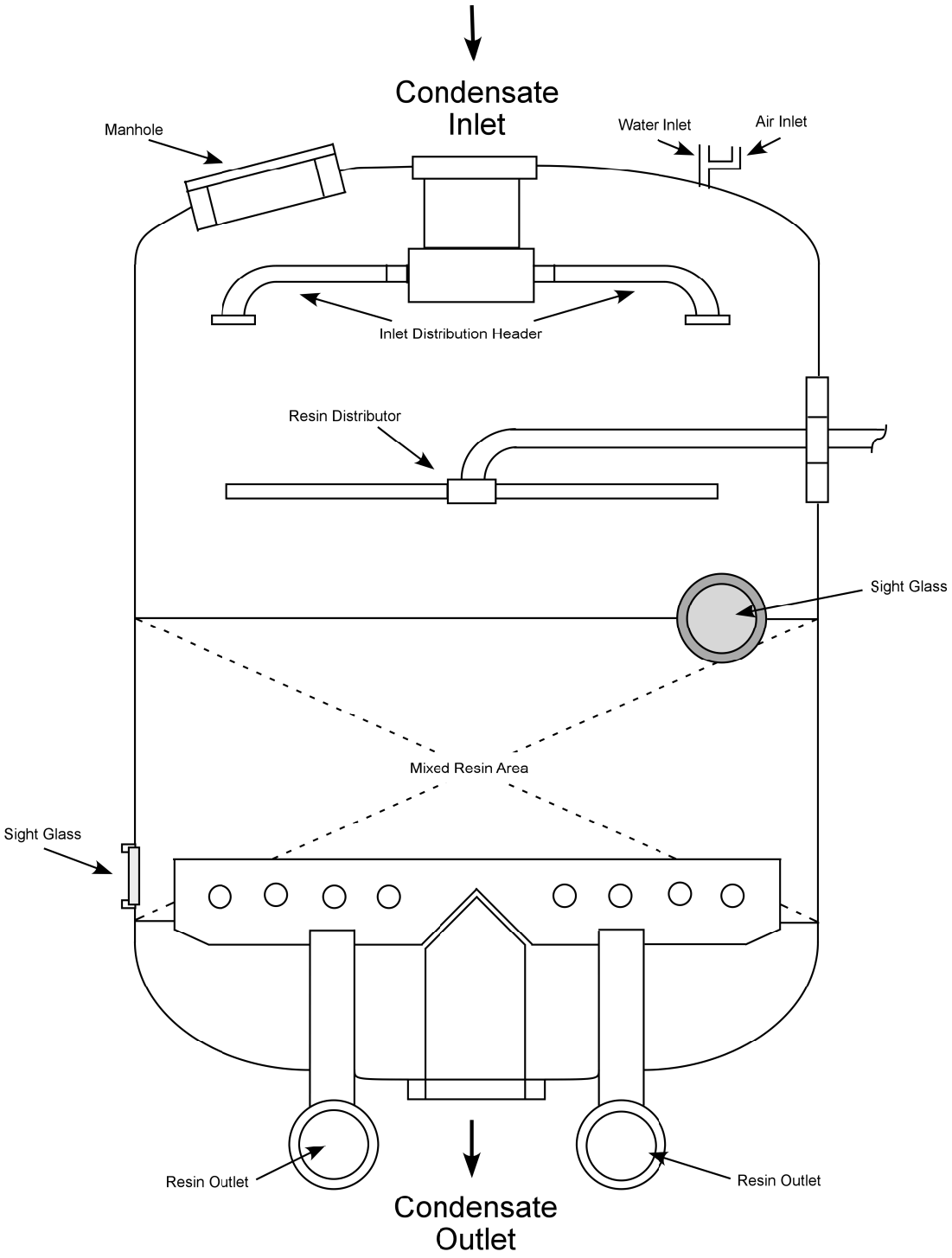
- Condense steam from first stage steam jet air ejectors
- Condensate flows through both SJAE condensers

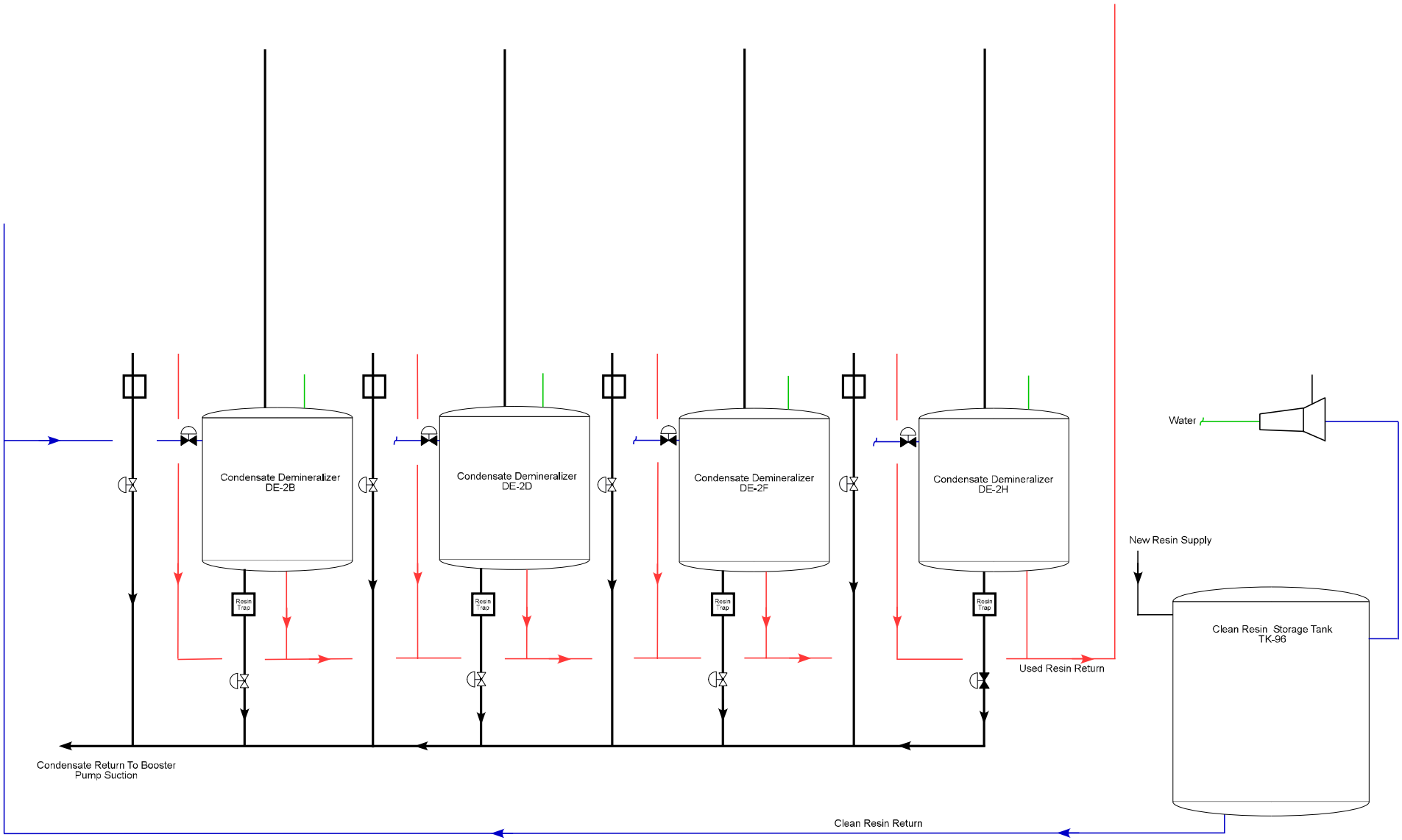
Steam Packing Exhauster

- Condenses leak-off steam from turbine seals
  - Main Turbine
  - Reactor Feed Pumps

Condensate Demineralizers

- Remove dissolved and suspended impurities
- Purify steam cycle water
- Maintain reactor water quality limits
- 8 units total, 7 required for full power
- Ultrasonic resin cleaning
- High conductivity or high D/P requires resin replacement


Condensate
Inlet
Manhole
Water Inlet
Air Inlet
Inlet Distribution Header
Resin Distributor
Sight Glass
Mixed Resin Area
Sight Glass
Resin Outlet
Resin Outlet
Condensate
Outlet

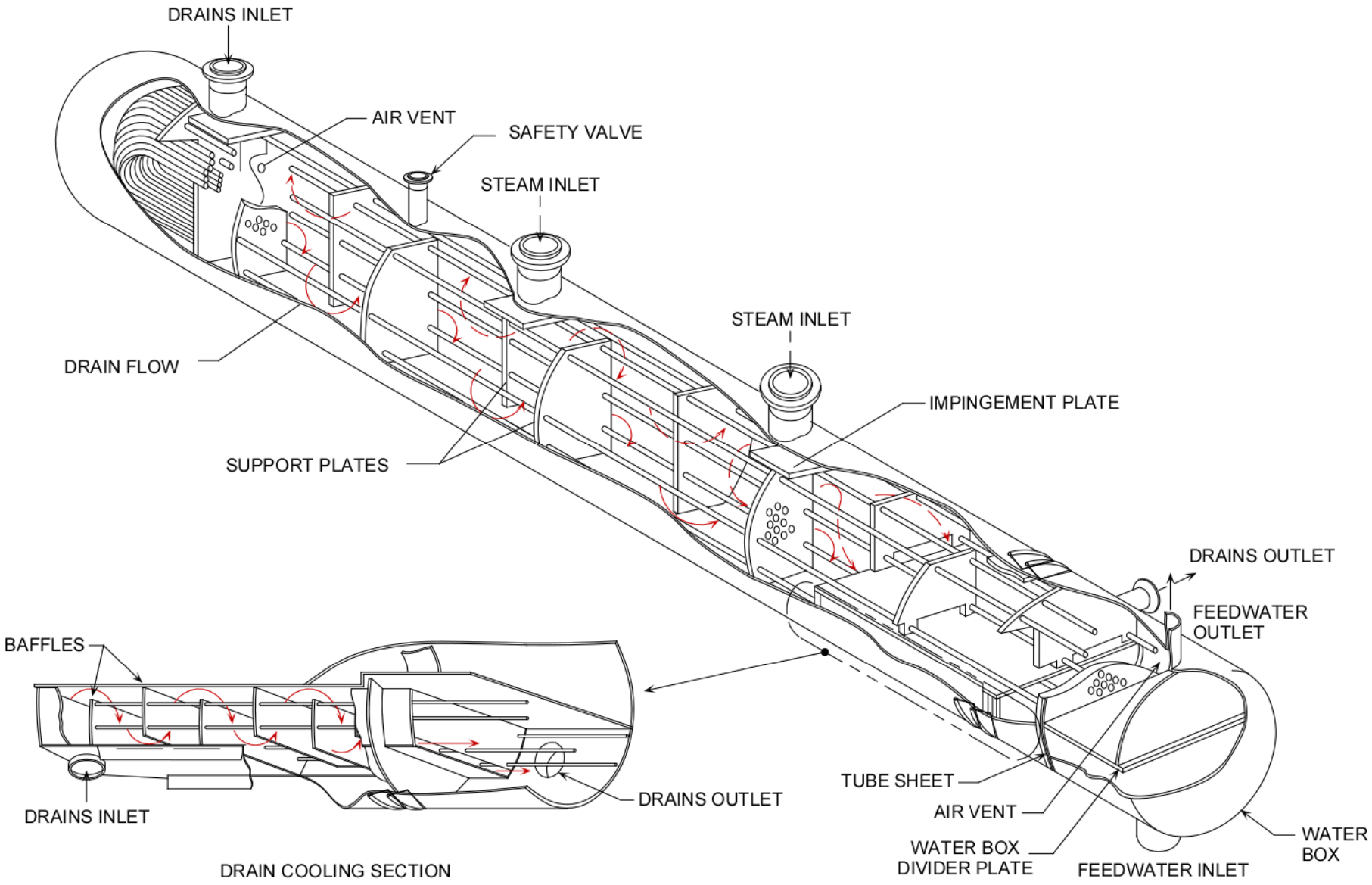

Condensate Demineralizer
DE-2B
Condensate Demineralizer
DE-2D
Condensate Demineralizer
DE-2F
Condensate Demineralizer
DE-2H
Resin Trap
Resin Trap
Resin Trap
Resin Trap
Water
New Resin Supply
Clean Resin Storage Tank
TK-96
Used Resin Return
Condensate Return To Booster
Pump Suction
Clean Resin Return

Condensate Booster Pumps

- Two pumps
- Provide adequate net positive suction head for Reactor Feed Pumps
- Capacity 12,153 gpm at ~ 600 psig

Low Pressure Heaters

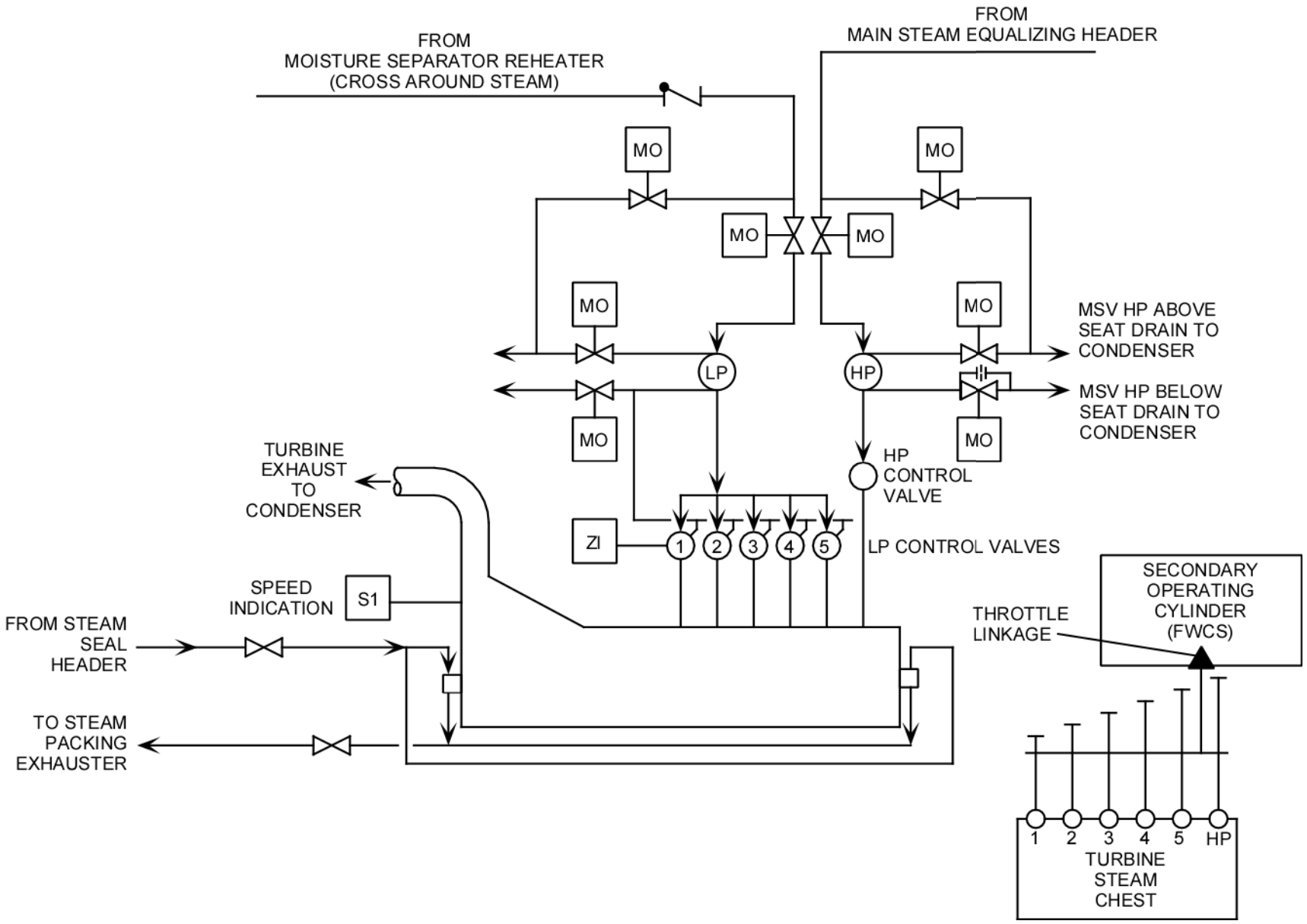
- Increase Plant Efficiency
- Two parallel heater string paths
- Extraction steam used to heat feedwater
- Feedwater flows through steam region on first pass
- Second pass through the water box portion of the heater
- Drains cascade to a lower pressure heater

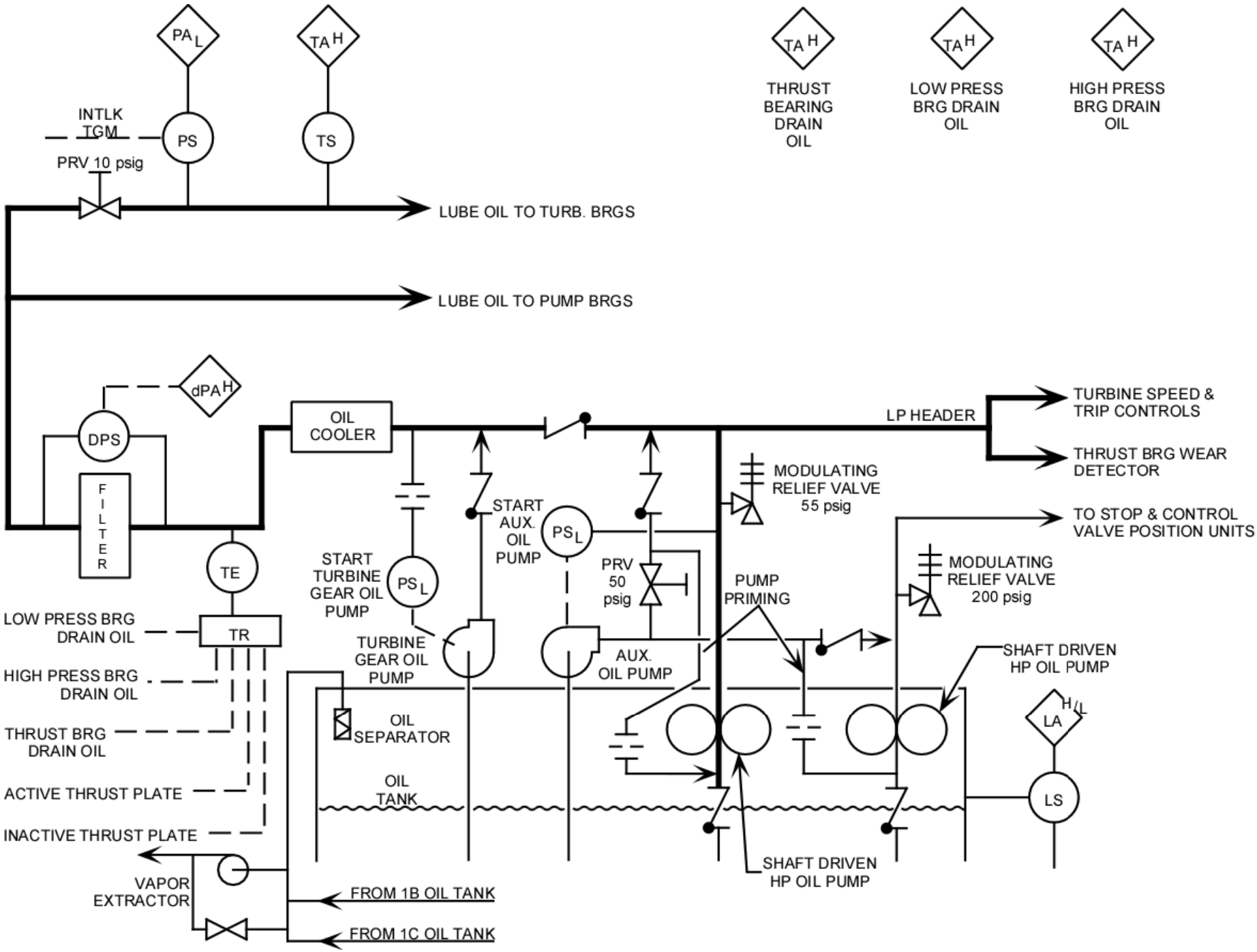


DRAIN COOLING SECTION

Feedwater Pumps

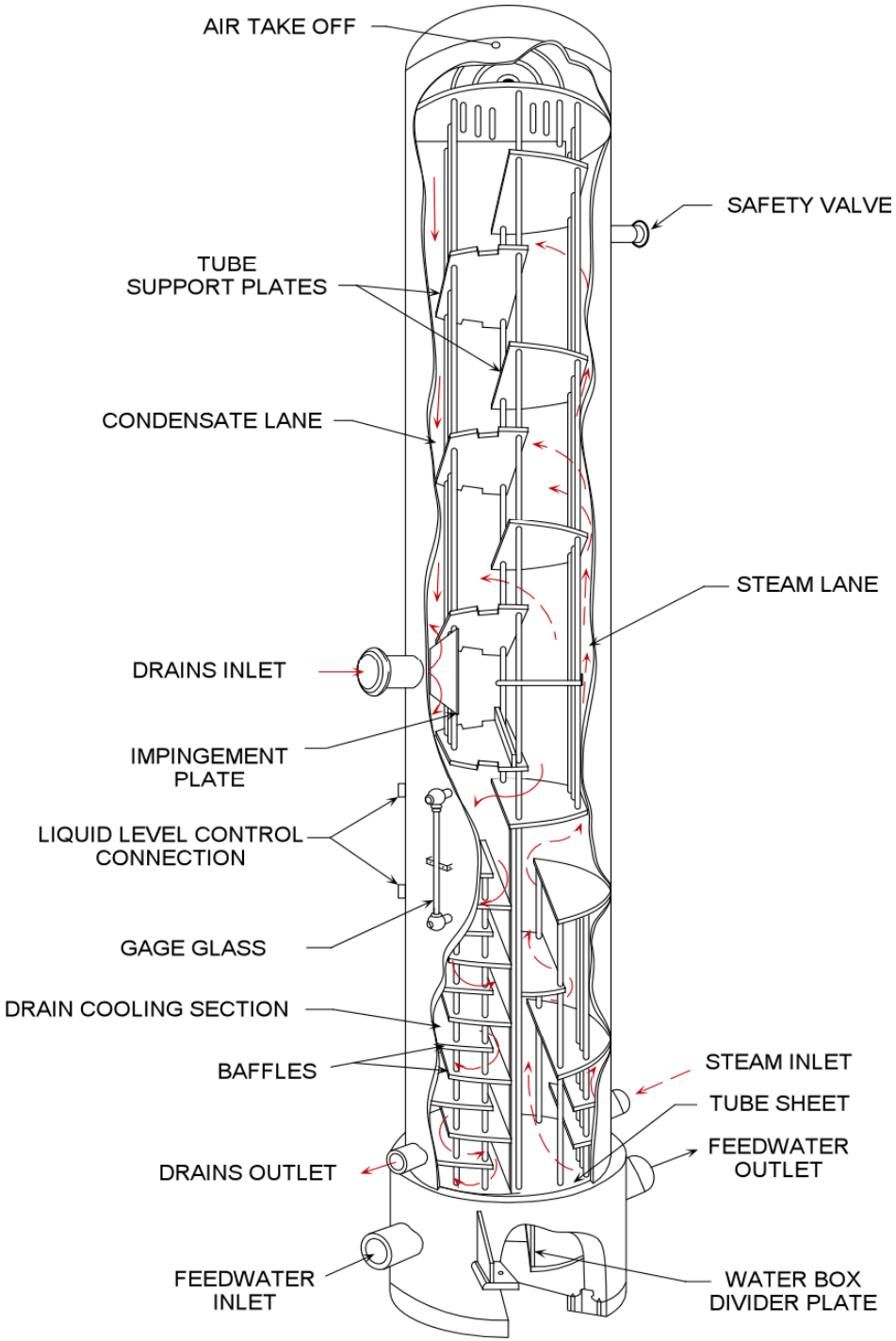
- Two steam driven pumps
- Provide driving force to inject water into reactor vessel
- 14,000 gpm at ~ 1130 psig
- 67% of system capacity each
- Two sets of control valves
  - Low pressure steam from Moisture Separator/Reheaters, normal supply
  - High pressure steam from Main Steam Equalizing Header

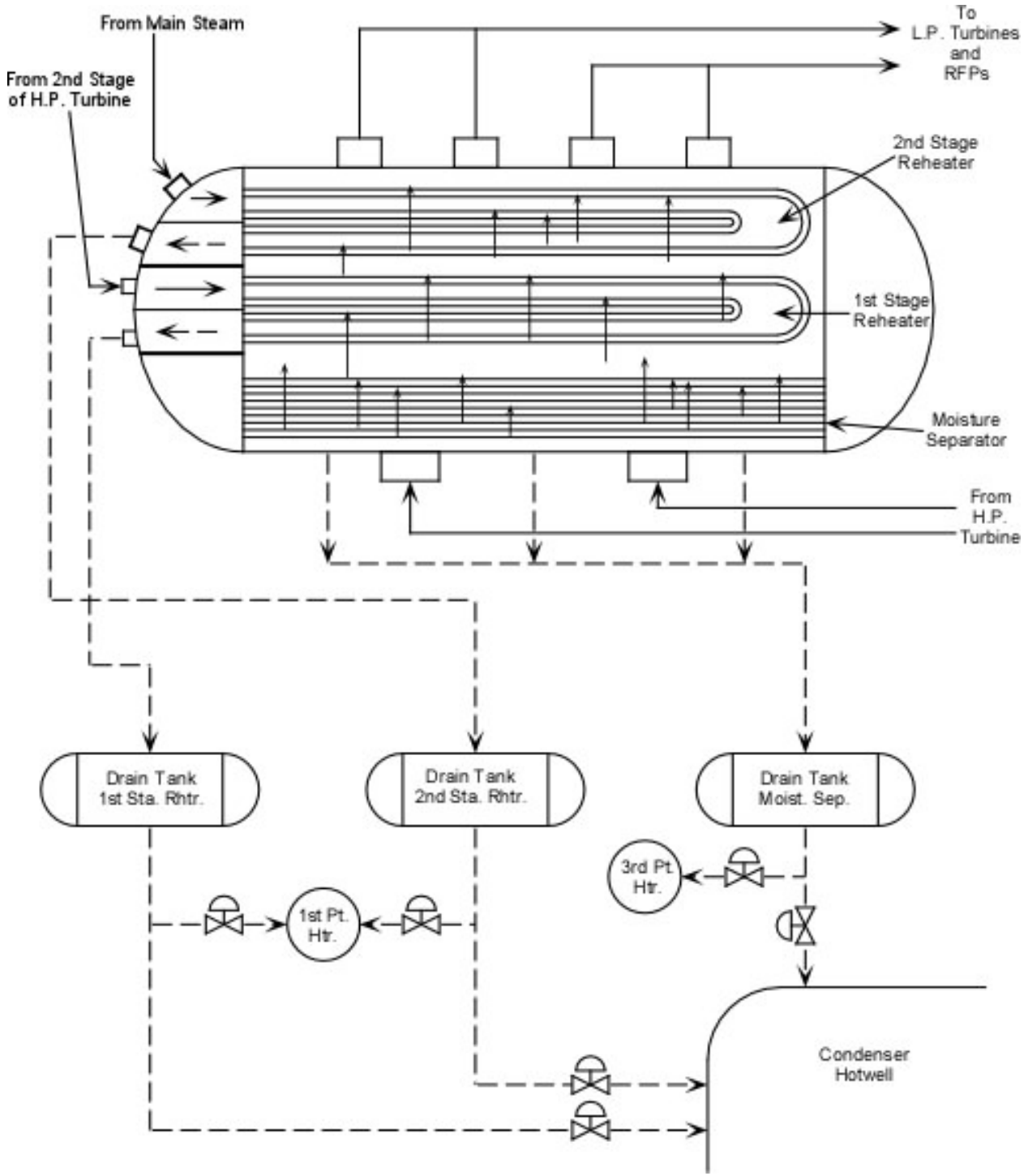

FROM
MOISTURE SEPARATOR REHEATER
(CROSS AROUND STEAM)
FROM
MAIN STEAM EQUALIZING HEADER
MO
MO
MO
MO
MO
MO
MO
MO
LP
HP
MSV HP ABOVE
SEAT DRAIN TO
CONDENSER
MSV HP BELOW
SEAT DRAIN TO
CONDENSER
TURBINE
EXHAUST
TO
CONDENSER
HP
CONTROL
VALVE
ZI
1
2
3
4
5
LP CONTROL VALVES
SPEED
INDICATION
S1
FROM STEAM
SEAL
HEADER
TO STEAM
PACKING
EXHAUSTER
THROTTLE
LINKAGE
SECONDARY
OPERATING
CYLINDER
(FWCS)
1
2
3
4
5
HP
TURBINE
STEAM
CHEST


PA L
TA H
INTLK TGM
PS
TS
PRV 10 psig
LUBE OIL TO TURB. BRGS
LUBE OIL TO PUMP BRGS
TA H
THRUST BEARING DRAIN OIL
TA H
LOW PRESS BRG DRAIN OIL
TA H
HIGH PRESS BRG DRAIN OIL
dPA H
DPS
FILTER
OIL COOLER
LP HEADER
TURBINE SPEED & TRIP CONTROLS
THRUST BRG WEAR DETECTOR
MODULATING RELIEF VALVE 55 psig
START AUX. OIL PUMP
PS L
TO STOP & CONTROL VALVE POSITION UNITS
TE
START TURBINE GEAR OIL PUMP
PS L
PRV 50 psig
PUMP PRIMING
MODULATING RELIEF VALVE 200 psig
LOW PRESS BRG DRAIN OIL
TR
TURBINE GEAR OIL PUMP
AUX. OIL PUMP
SHAFT DRIVEN HP OIL PUMP
HIGH PRESS BRG DRAIN OIL
OIL SEPARATOR
H L
LA
THRUST BRG DRAIN OIL
OIL TANK
ACTIVE THRUST PLATE
LS
INACTIVE THRUST PLATE
SHAFT DRIVEN HP OIL PUMP
VAPOR EXTRACTOR
FROM 1B OIL TANK
FROM 1C OIL TANK

High Pressure Heaters

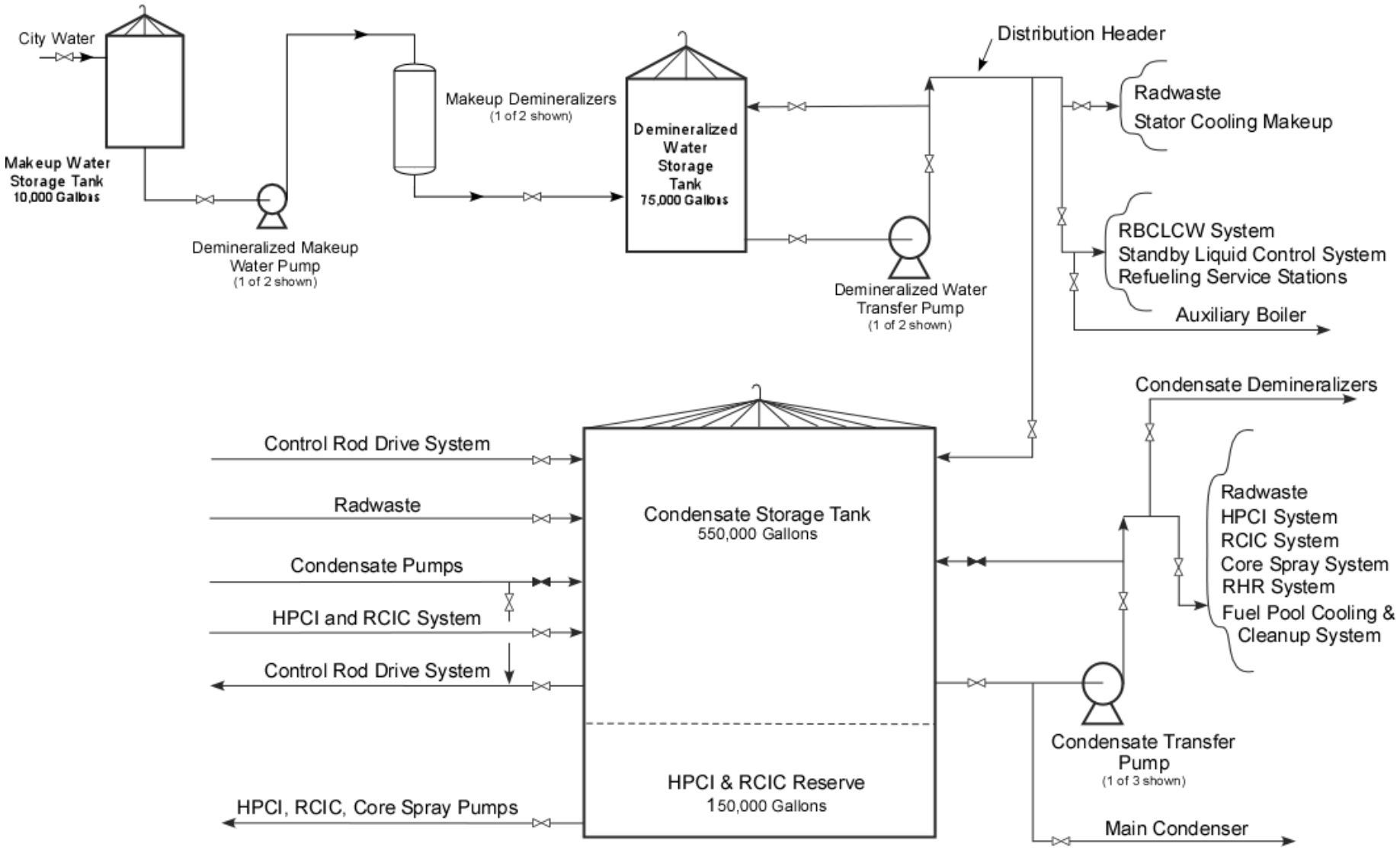
- Last stage of feedwater heating
- Uses high pressure extraction steam from HP Turbine
- ~ 420°F Feedwater Injection temperature to reactor vessel


AIR TAKE OFF
SAFETY VALVE
TUBE
SUPPORT PLATES
CONDENSATE LANE
STEAM LANE
DRAINS INLET
IMPINGEMENT
PLATE
LIQUID LEVEL CONTROL
CONNECTION
GAGE GLASS
DRAIN COOLING SECTION
BAFFLES
STEAM INLET
TUBE SHEET
FEEDWATER
OUTLET
DRAINS OUTLET
FEEDWATER
INLET
WATER BOX
DIVIDER PLATE

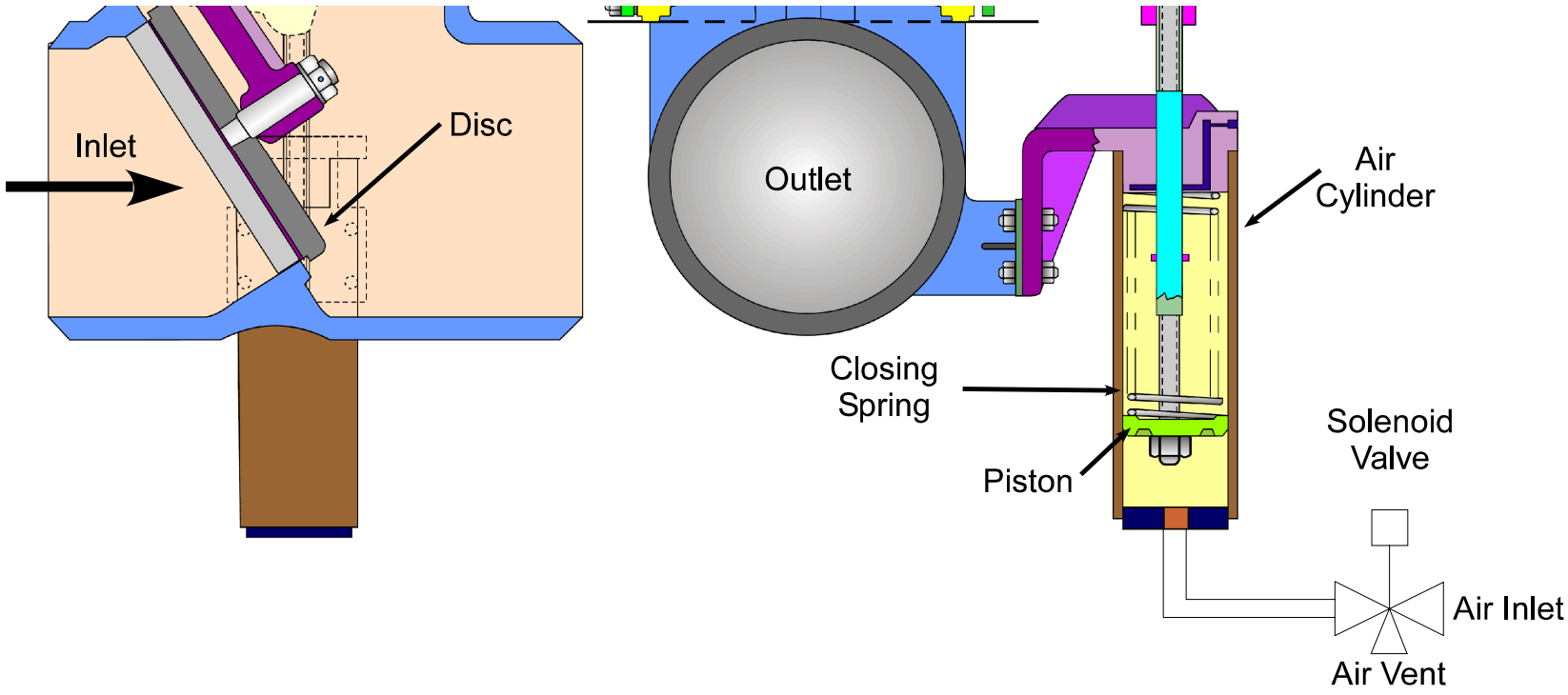

From Main Steam
From 2nd Stage of H.P. Turbine
To L.P. Turbines and RFPs
2nd Stage Reheater
1st Stage Reheater
Moisture Separator
From H.P. Turbine
Drain Tank 1st Sta. Rhtr.
Drain Tank 2nd Sta. Rhtr.
Drain Tank Moist. Sep.
3rd Pt. Htr.
1st Pt. Htr.
Condenser Hotwell

Condensate Storage Tank

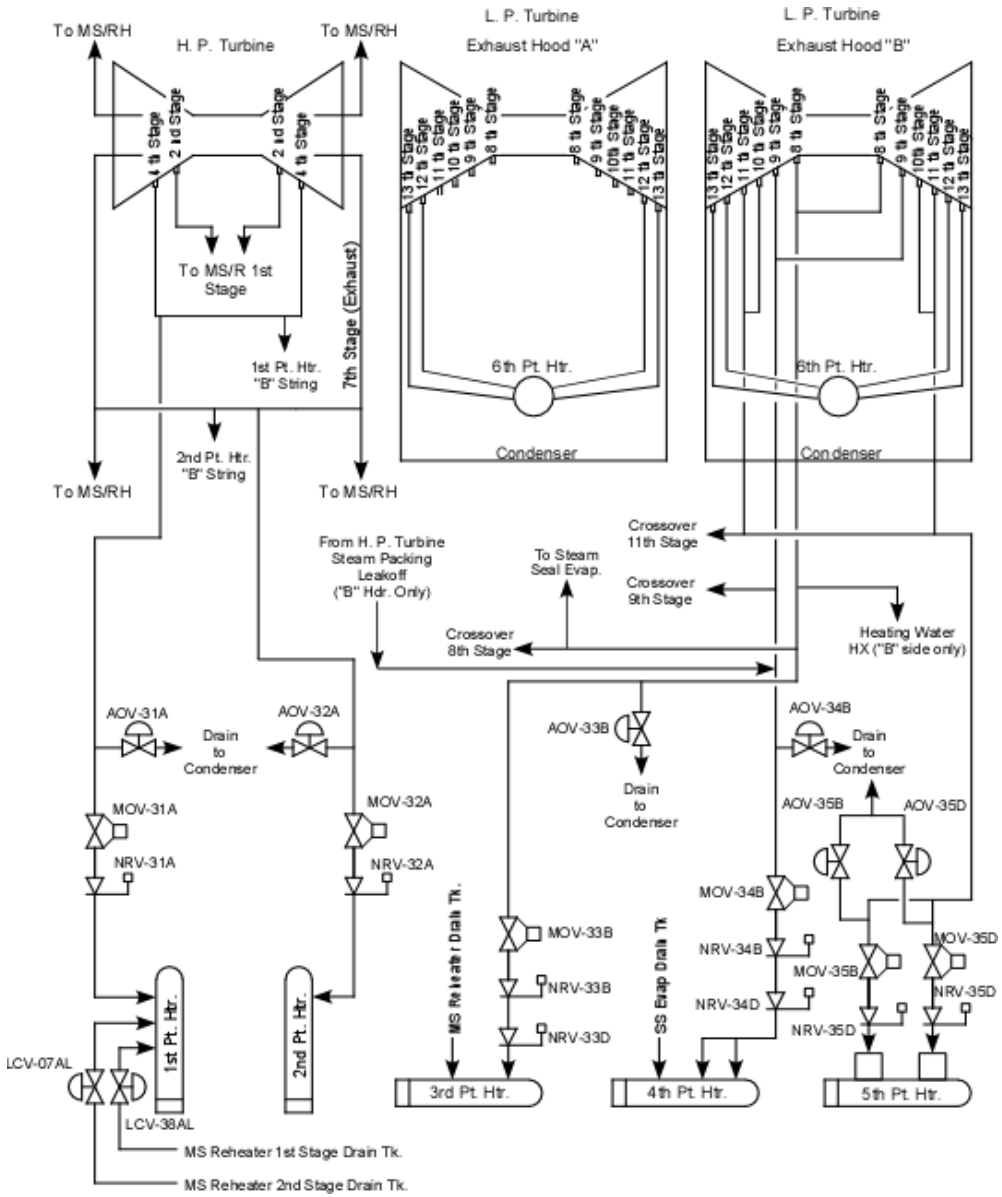
- Capacity - ~ 500,000 gallons
- 150,000 gallons dedicated for
  - High Pressure Coolant Injection
  - Reactor Core Isolation Cooling
- Typically located outdoors
- Provides makeup supply to hotwell
- Provides rejection source for hotwell


City Water
Makeup Water Storage Tank 10,000 Gallons
Demineralized Makeup Water Pump (1 of 2 shown)
Makeup Demineralizers (1 of 2 shown)
Demineralized Water Storage Tank 75,000 Gallons
Demineralized Water Transfer Pump (1 of 2 shown)
Distribution Header
Radwaste
Stator Cooling Makeup
RBCLCW System
Standby Liquid Control System
Refueling Service Stations
Auxiliary Boiler
Condensate Demineralizers
Radwaste
HPCI System
RCIC System
Core Spray System
RHR System
Fuel Pool Cooling & Cleanup System
Control Rod Drive System
Radwaste
Condensate Pumps
HPCI and RCIC System
Control Rod Drive System
Condensate Storage Tank 550,000 Gallons
HPCI & RCIC Reserve 150,000 Gallons
HPCI, RCIC, Core Spray Pumps
Condensate Transfer Pump (1 of 3 shown)
Main Condenser

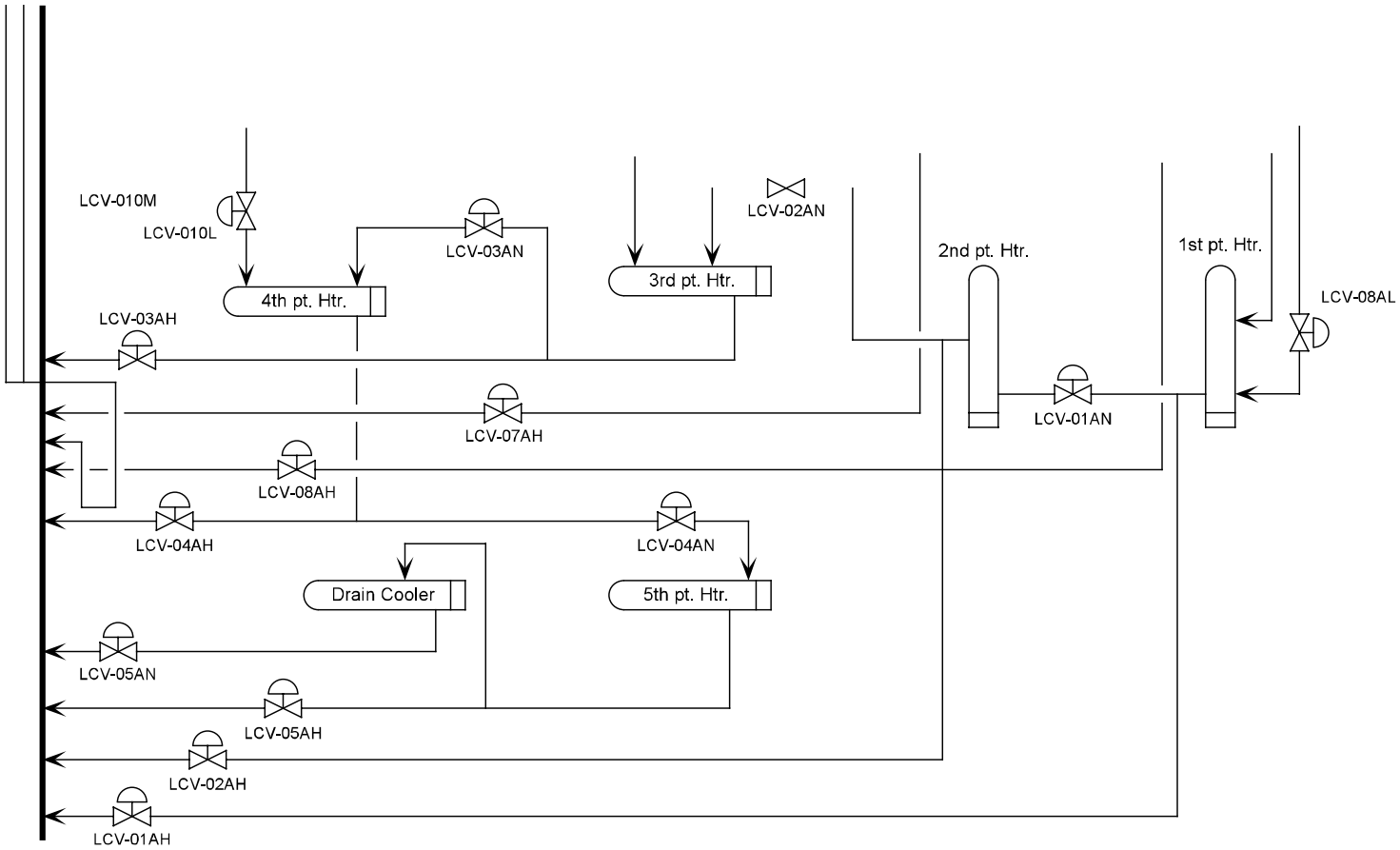
Non-Return Valve



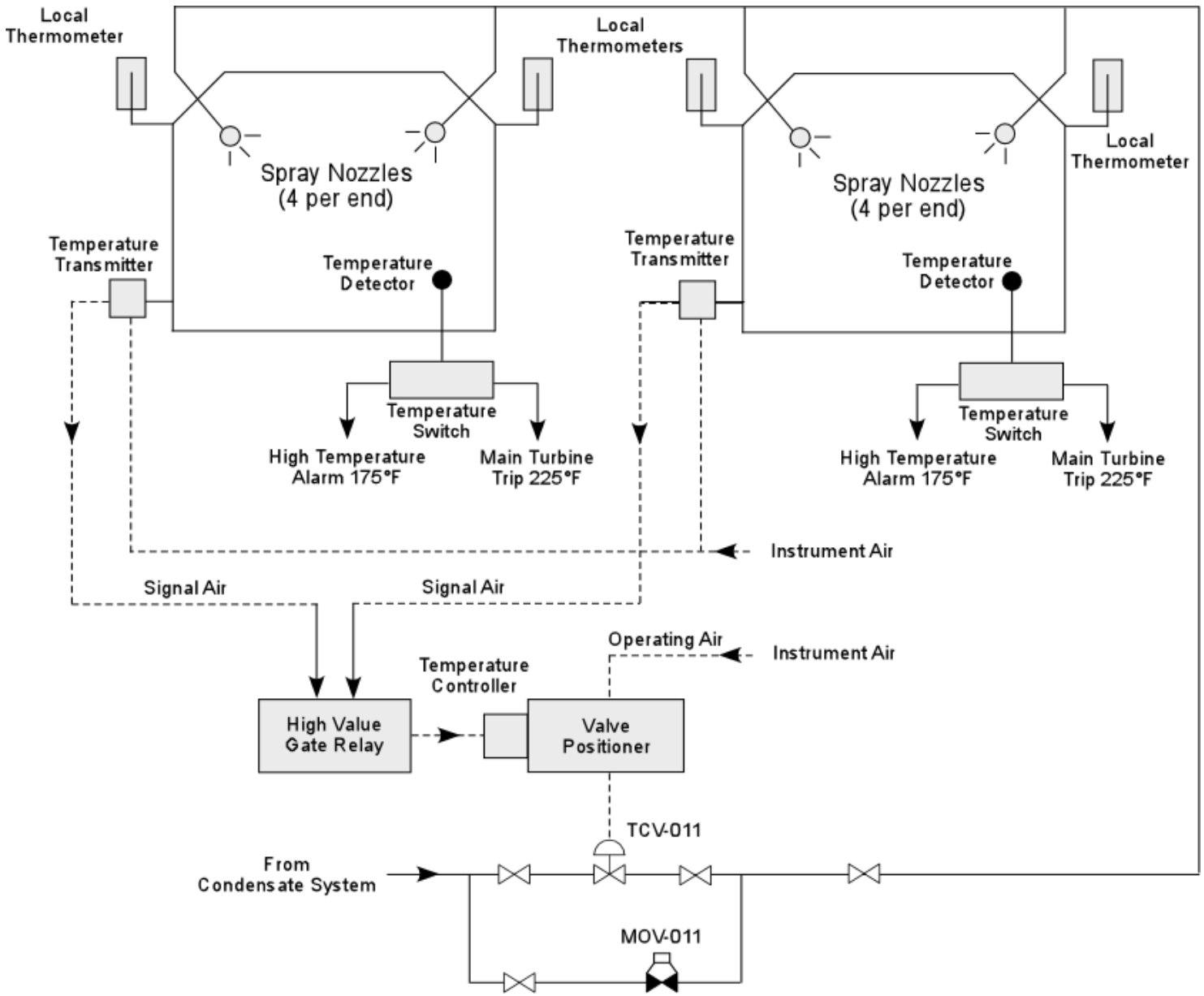
Extraction Steam



Heater Drains

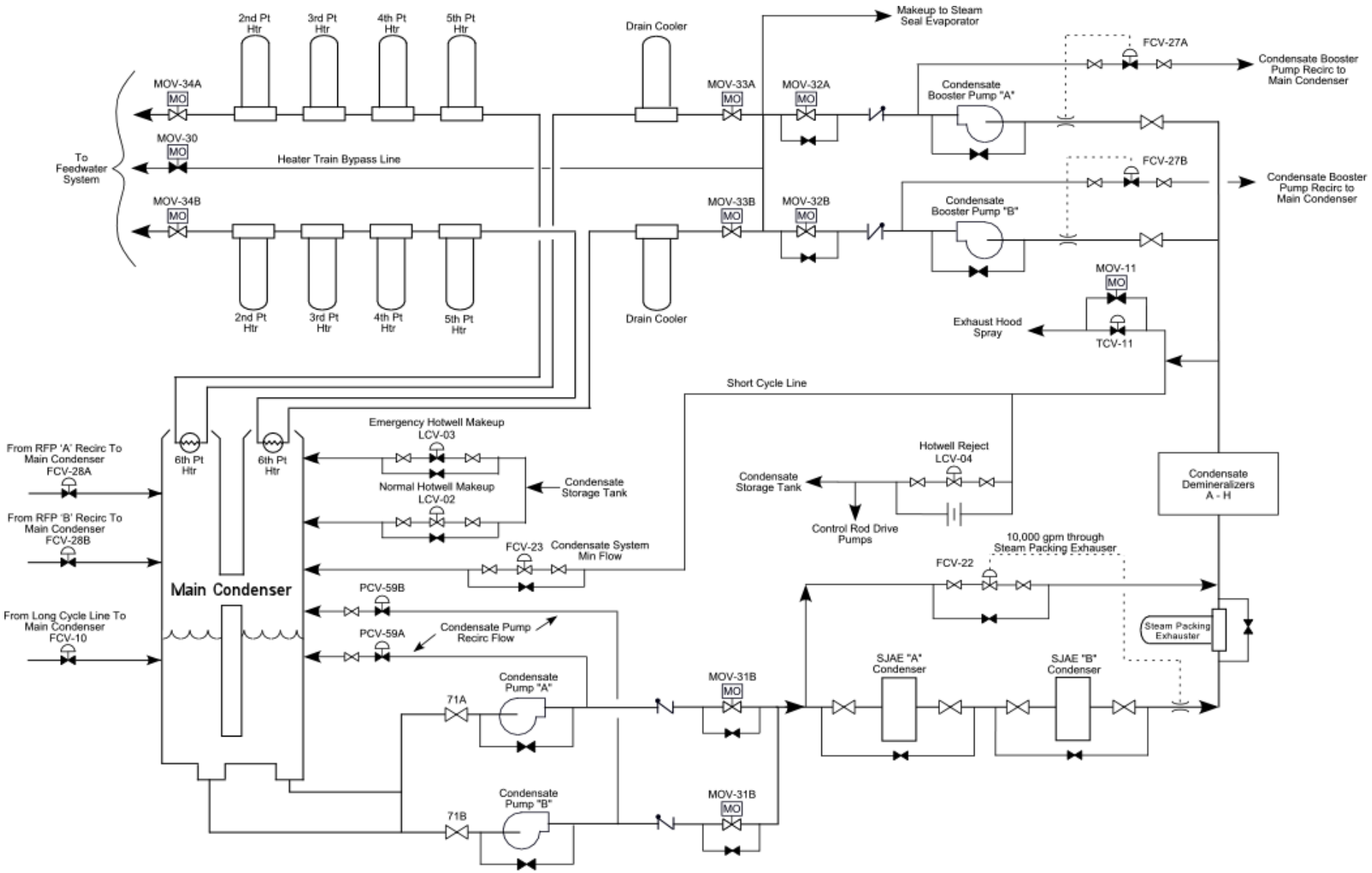


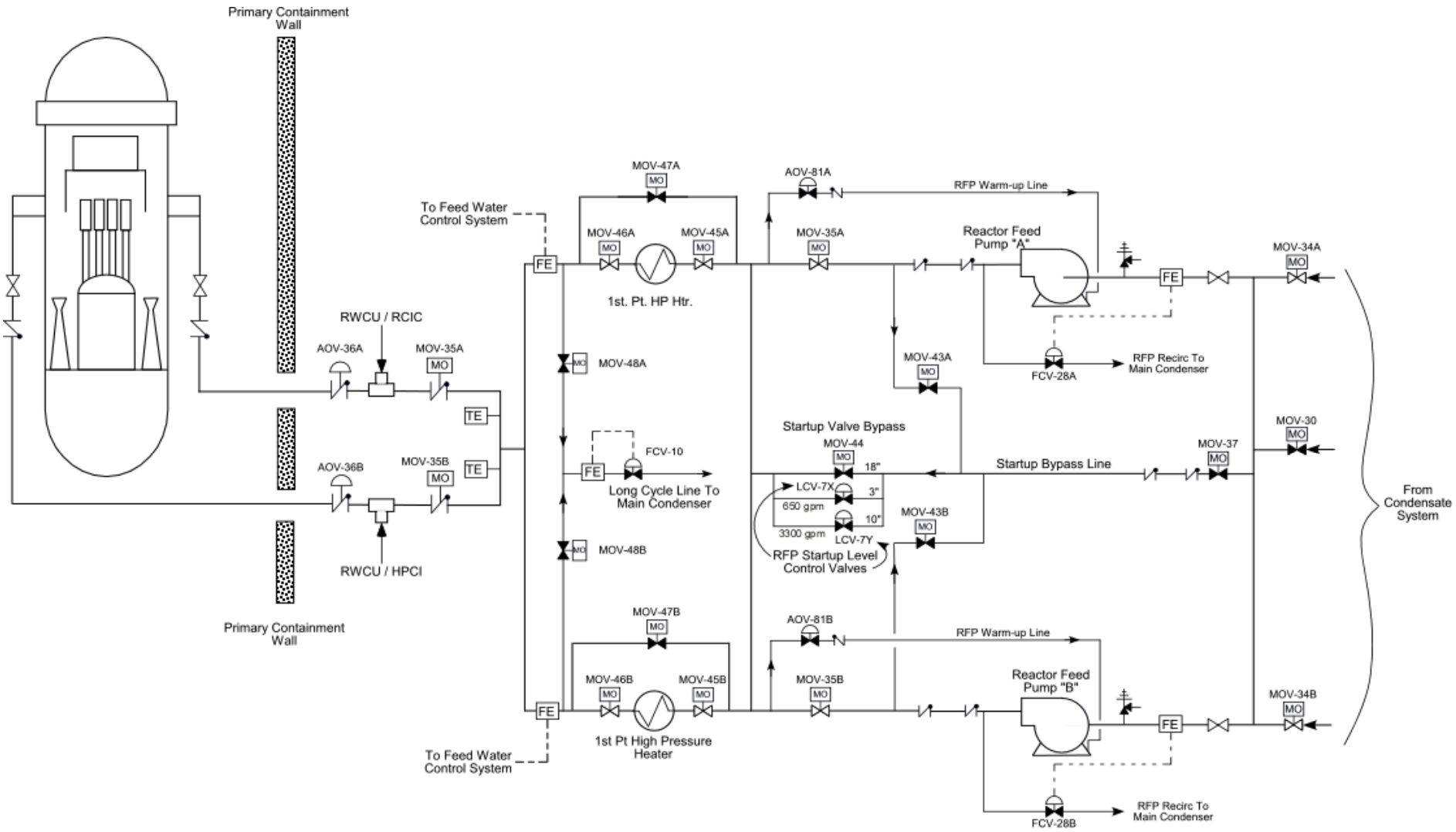
Exhaust Hood Spray

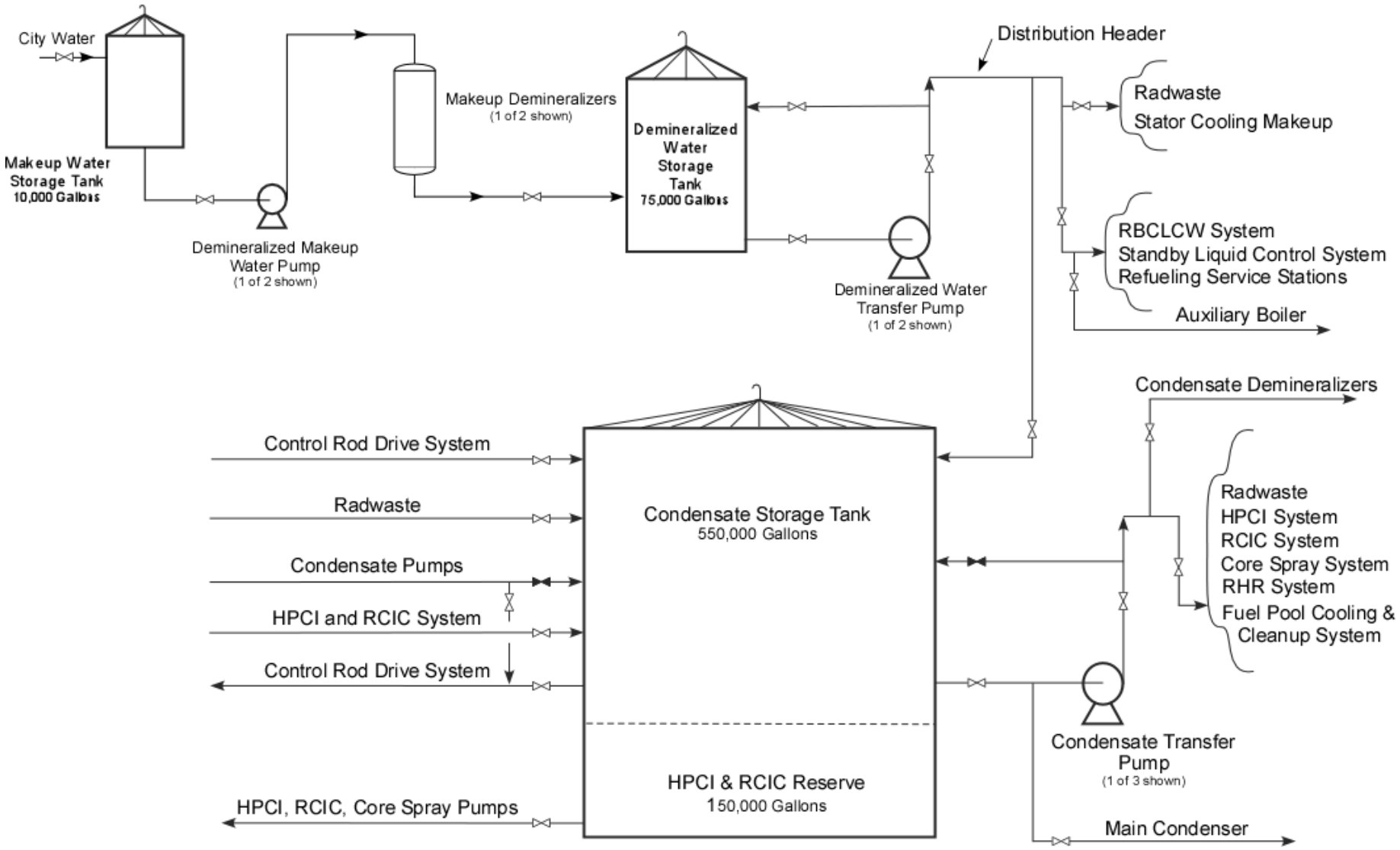


Condensate & Feedwater System Flowpaths

- Hotwell Level Control
- Condensate System
- Feedwater System
- Exhaust Hood Spray
- Short Cycle Cleanup
- Long Cycle Cleanup
- Demineralizer Water System
- Condensate Storage & Transfer System


2nd Pt Htr
3rd Pt Htr
4th Pt Htr
5th Pt Htr
Drain Cooler
Makeup to Steam Seal Evaporator
FCV-27A
Condensate Booster Pump Recirc to Main Condenser
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MOV-32A
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To Feedwater System
Heater Train Bypass Line
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From RFP 'A' Recirc To Main Condenser FCV-28A
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From Long Cycle Line To Main Condenser FCV-10
6th Pt Htr
Main Condenser
FCV-23
Condensate System Min Flow
Control Rod Drive Pumps
10,000 gpm through Steam Packing Exhauser
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PCV-59B
PCV-59A
Condensate Pump Recirc Flow
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SJAE "A" Condenser
SJAE "B" Condenser
Condensate Pump "A"
71A
MOV-31B
Condensate Pump "B"
71B


Primary Containment Wall
RWCU / RCIC
AOV-36A
MOV-35A
MO
TE
AOV-36B
MOV-35B
RWCU / HPCI
Primary Containment Wall
To Feed Water Control System
FE
MOV-47A
MOV-46A
MOV-45A
1st. Pt. HP Htr.
MOV-48A
FCV-10
Long Cycle Line To Main Condenser
MOV-48B
MOV-47B
MOV-46B
MOV-45B
1st Pt High Pressure Heater
To Feed Water Control System
AOV-81A
RFP Warm-up Line
MOV-35A
Reactor Feed Pump "A"
MOV-43A
FCV-28A
RFP Recirc To Main Condenser
Startup Valve Bypass
MOV-44
18"
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3"
650 gpm
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From Condensate System
AOV-81B
RFP Warm-up Line
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Reactor Feed Pump "B"
FCV-28B
RFP Recirc To Main Condenser
MOV-34B


City Water
Makeup Water Storage Tank 10,000 Gallons
Demineralized Makeup Water Pump (1 of 2 shown)
Makeup Demineralizers (1 of 2 shown)
Demineralized Water Storage Tank 75,000 Gallons
Demineralized Water Transfer Pump (1 of 2 shown)
Distribution Header
Radwaste
Stator Cooling Makeup
RBCLCW System
Standby Liquid Control System
Refueling Service Stations
Auxiliary Boiler
Condensate Demineralizers
Radwaste
HPCI System
RCIC System
Core Spray System
RHR System
Fuel Pool Cooling & Cleanup System
Control Rod Drive System
Radwaste
Condensate Pumps
HPCI and RCIC System
Control Rod Drive System
Condensate Storage Tank 550,000 Gallons
HPCI & RCIC Reserve 150,000 Gallons
HPCI, RCIC, Core Spray Pumps
Condensate Transfer Pump (1 of 3 shown)
Main Condenser

Condensate & Feedwater System Setpoints

- Condensate Demineralizer High Temperature
- Condensate Demineralizer Resin Depletion
- Condensate Booster Pump Low Suction Pressure
- Feedwater Pump Low Suction Pressure

Demineralizer High Temperature

- $>$ 130°F
- Manually bypass
- Resin melts ~ 150°F to 160°F

Demineralizer Resin Depletion

- High Differential Pressure – 40 psig
  - Typically requires cleaning
- 0.1 µmho/cm
  - Typically requires resin replacement

Suction Pressure Trips

- Condensate Booster Pump Low Suction
  - 35 psig
  - ‘A’ Pump time delay trip 20 seconds
  - ‘B’ Pump time delay trip 45 seconds
- Feedwater Pump Low Suction
  - 250 psig
  - ‘A’ Pump time delay trip 8 seconds
  - ‘B’ Pump time delay trip 25 seconds

System Interrelations

- Circulating Water System
- Control Rod Drive System
- Feedwater Level Control System
- Reactor Water Cleanup System
- High Pressure Coolant Injection System
- Reactor Core Isolation Cooling System
- Main Steam System
- Offgas System

Objective Review

1. Identify the purposes of the Condensate and Feedwater system.
2. Recognize the purpose, function and operation of the following Condensate and Feedwater system major components:
    - Condensate Storage Tank
    - Main Condenser
    - Main Condenser Hotwell
    - Condensate Pumps
    - Steam Jet Air Ejector Condenser
    - Steam Packing Exhauster
    - Condensate Demineralizers
    - Condensate Booster Pumps
    - Low Pressure Heaters
    - Feedwater Pumps
    - High Pressure Heaters

Objective Review

3. Describe the following flowpaths of the Condensate and Feedwater system:
    - Demineralizer Water System
    - Condensate Storage and Transfer System
    - Hotwell Level Control
    - Condensate System
    - Feedwater System
    - Exhaust Hood Spray
    - Short Cycle Cleanup
    - Long Cycle Cleanup

Objective Review

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    - Offgas System

Questions
- or –
Comments